QuantForge

Automated Algorithm Discovery through Evidence-Guided Program Evolution

HYPOTHESIZE · EXPERIMENT · LEARN · REVISE

"Separate what works from why it works."

From experiments to better algorithms.

Frozen Task Contract

- Protocol
- Data/Env

Candidate Program

Initial solution

Programs *Evolve programs using evidence.*

Current Program P_t

Revised Program P_{t+1}

Retained Program P^*

scores

Performance Archive

Programs, results, metadata

under test

guide revision

Experiment Execution

Run targeted tests

results

Residual Compilation

Compile verdicts into constraints

Discovered Algorithm

Best program with research trace

Evidence *Refine empirical error models.*

Residual Hypothesis

Explain remaining error

Discriminative Test

Targeted comparison

Supported / Refuted Residual

Update error model

Evidence Record

Tests, outcomes, interpretations

AUTOMATED SCIENTIFIC DISCOVERY

Programs and error models co-evolve through experiment.

EVIDENCE DRIVES PROGRESS